Parametric Maximum Likelihood Estimation for Locally Stationary Hidden Markov Models

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Structure

- What is a Hidden Markov model?
- What does *locally stationary* mean?
- Adaptation of a classical consistency proof?
- The AR(1) process and L^p-local stationarity.

What is a Hidden Markov Model?

An introductory example

- Imagine **A**lice and **B**ob use a **C**hannel for signal transmission.
- **A** sends signals $(X_0, \ldots, X_{n-1})$ where X_k takes values in $\{0, 1\}$.

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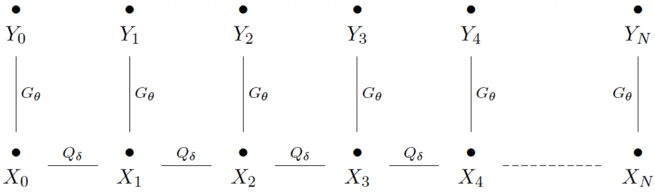
- **B** wants to find out how broken the channel is.
- The array of signals will be assumed to be a Markov chain, i. e. $\mathrm{P}(X_k = i | X_{k-1}, \dots, X_0) = \mathrm{P}(X_k = i | X_{k-1})$
- Transition probabilities $Q_\delta(j, \{i\}) = \mathrm{P}_\delta(X_k = i | X_{k-1} = j)$ are known.

Situation can be mathematically modeled in the following way.

- Consider $(\mathcal{X} := \{0, 1\}, \mathcal{B}(\mathcal{X}))$ and $(\mathcal{Y} := \{0, 1\}, \mathcal{B}(\mathcal{Y}))$.
- (X_k, Y_k) is a bivariate Markov chain on $\mathcal{X} \times \mathcal{Y}$ with the following dependence structure.



Say the kernels are represented by the matrices

$$Q_\delta = \begin{pmatrix} Q_\delta(0,0) & Q_\delta(1,0) \\ Q_\delta(0,1) & Q_\delta(1,1) \end{pmatrix} = \begin{pmatrix} 1-\delta & \delta \\ \delta & 1-\delta \end{pmatrix} \text{ for } \delta \in (0,1)$$

$$G_\theta = \begin{pmatrix} G_\theta(0,0) & G_\theta(1,0) \\ G_\theta(0,1) & G_\theta(1,1) \end{pmatrix} = \begin{pmatrix} 1-\theta & \theta \\ \theta & 1-\theta \end{pmatrix} \text{ for } \theta \in [\delta, 1-\delta] =: \Theta.$$

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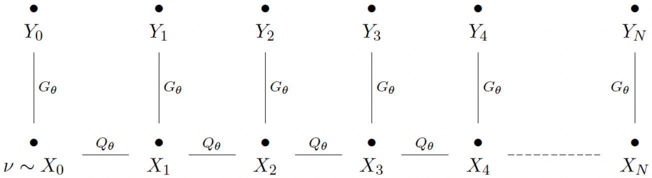
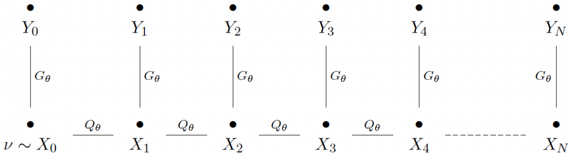
- **B** knows δ and could now resort to the highly developed theory of *Hidden Markov models* – for example Maximum likelihood estimation for $\theta \in [\delta, 1-\delta]$.
- Hidden Markov Models are models of the above form. In the *classical and well-understood* homogeneous case



The formal definition

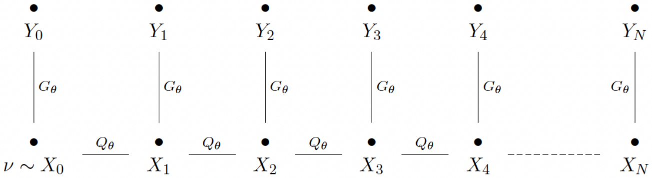


Definition (Hidden Markov Models)

Let $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ and $(\mathcal{Y}, \mathcal{A}_\mathcal{Y})$ be measurable spaces and let $(Z_k)_k = ((X_k, Y_k))_k$ be a bivariate Markov chain on $\mathcal{X} \times \mathcal{Y}$. We say that $(Z_k)_k$ is a Hidden Markov Model, if for every k there is a Markov kernel Q_k from $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ to itself and a Markov kernel G_k from $(\mathcal{X}, \mathcal{A}_\mathcal{X})$ to $(\mathcal{Y}, \mathcal{A}_\mathcal{Y})$ such that for every bounded and measurable $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ we have

$$\mathrm{E}\left[f(X_k, Y_k) | X_0^{k-1}, Y_0^{k-1}\right] = \int_\mathcal{X} \int_\mathcal{Y} f(x, y) G_k(x, dy) Q_k(X_{k-1}, dx)$$

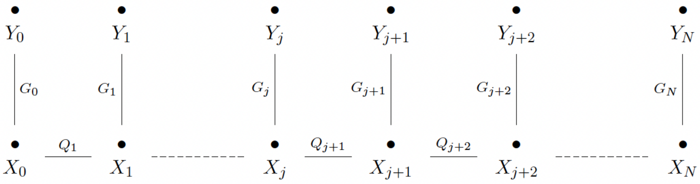
Applications of Hidden Markov Models



- Real-world phenomena X_k that are only observed via noisy or incomplete data.
  - physical quantities observed in experiments
  - evolution of medical indicators whose measurements are noisy
  - ...
- live-tracking of moving objects
- ...

What is local stationarity?

The issue of homogeneity and stationarity



- Homogeneity/Stationarity is a key requirement for the general consistency results on maximum likelihood estimators to hold.

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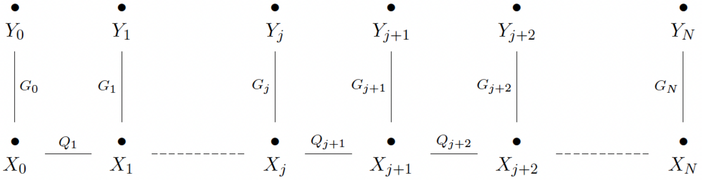

- Homogeneity/Stationarity is a key requirement for the general consistency results on maximum likelihood estimators to hold.
- … and also to somewhat even make sense.
- $\hat{\theta}_N \in \operatorname{argmax}_{\theta \in \Theta} \log p_\theta(Y_0^N)$ not expected to approximate anything useful, if transition parameters evolve *drastically* over time.

Typical situation:

- Provided with an array of $N+1$ many observations $Y_0, \ldots, Y_N$
- and reasonable to assume transition kernels Q and G are approximately the same around $\lfloor tN \rfloor$ for $\approx bN$ many observations.
- Aim to find out what the kernels around $\lfloor tN \rfloor$ look like.

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$$
\begin{tikzcd}
Y_0 \arrow[d, dash, "G_0"'] & Y_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor} \arrow[d, dash, "G_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor}"'] \arrow[rr, bend left, dashed, dash, "\approx G_{\lfloor tN \rfloor}"] & Y_{\lfloor tN \rfloor} \arrow[d, dash, "G_{\lfloor tN \rfloor}"'] & Y_{\lfloor tN \rfloor + \lfloor bN \rfloor/2} \arrow[d, dash, "G_{\lfloor tN \rfloor + \lfloor bN/2 \rfloor}"'] & Y_N \arrow[d, dash, "G_N"'] \\
X_0 \arrow[r, dashed, dash, "Q_*" description] & X_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor} \arrow[r, dashed, dash] \arrow[rr, bend right, dashed, dash, "\approx Q_{\lfloor tN \rfloor}"'] & X_{\lfloor tN \rfloor} \arrow[r, dashed, dash] & X_{\lfloor tN \rfloor + \lfloor bN \rfloor/2} \arrow[r, dashed, dash, "Q_*" description] & X_N
\end{tikzcd}
$$

- Pretend homogeneity around $\lfloor tN \rfloor$
- and do maximum likelihood estimation *anyway* only including the observations from that window of size $\approx bN$

In conclusion, consider a maximum likelihood estimator resembling

$$\hat{\theta}_N \in \operatorname{argmax}_{\theta \in \Theta} \log \rho_\theta (Y_{\lfloor tN \rfloor - \lfloor bN/2 \rfloor}^{\lfloor tN \rfloor + \lfloor bN/2 \rfloor})$$

and hope $Q_{\hat{\theta}_N} \approx Q_k$ and $G_{\hat{\theta}_N} \approx G_k$ for

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- In our situation, including more data from $Y_0, \ldots, Y_N$ is **not an option**.

Developing an adequate asymptotic

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- Good would be an asymptotic $N \to \infty$, $b \to 0$ but still $bN \to \infty$.
- Effectively, we then estimate a parameter at a point $t \in [0, 1]$ in *unit time*.
- We get **both** more stationary around every $t \in [0, 1]$ and gather more observations as $N \to \infty$

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Definition (First (heuristic) definition for local stationarity)

- A triangular scheme $\{\{Z_{k,N}\}_{0\le k\le N}\}_{N\ge 0}$ of processes with finite dimensional distributions $\pi_{k,N}^{m,N}$
- And for every $t \in [0,1]$ a stationary process $(\widetilde{Z}_k(t))_{k\in\mathbb{Z}}$ with finite dimensional distributions $\pi_k^m(t) = \pi_{m-k+1}(t)$
- $\limsup_{N\to\infty} \sup_{0\le k\le m\le N, |m-k|\le l} \|\pi_k^m(k/N) - \pi_{k,N}^{m,N}\|_{TV} = 0 \qquad \forall l \ge 1$

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Of course, variations possible

- Other metric on the distributions' space possible
- And definition involves subtleties (continuity of limit distributions)
- Often possible to derive explicit bounds of the form

$$\|\pi_k^m(u) - \pi_{k,N}^{m,N}\|_{\mathrm{TV}} \le C_l \left[\left|u - \frac{k}{N}\right| + \frac{1}{N}\right] \text{ for } 0 \le m-k \le l$$

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What does the MLE look like now? Morally at least like

$$\hat{\theta}_N(t) \in \mathrm{argmax}_{\theta\in\Theta} \log \rho_\theta(Y_{\lfloor tN\rfloor - \lfloor bN/2\rfloor, N}^{\lfloor tN\rfloor + \lfloor bN/2\rfloor, N})$$

What happens as $N \to \infty$, $b \to 0$ but yet still $bN \to \infty$?

How can we adapt a consistency proof?

The classical case

First let us go back to **assuming stationarity**.

- In the classical iid case we decompose

$$n^{-1} \log \rho_\theta(X_0^{n-1}) = n^{-1} \sum_{k=0}^{n-1} \log \rho_\theta(X_k) \stackrel{n\to\infty}{\longrightarrow} \mathrm{E}\left[\log \rho_\theta(X_0)\right] =: \ell(\theta)$$

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- In Hidden Markov case it is harder.

$$\begin{aligned} n^{-1} \log p_\theta(Y_0^{n-1}) &= n^{-1} \sum_{k=0}^{n-1} \log \left(p_\theta(Y_0^k) / p_\theta(Y_0^{k-1}) \right) \\ &= n^{-1} \sum_{k=0}^{n-1} \log p_\theta(Y_k | Y_0^{k-1}) \end{aligned}$$

The likelihood function's (slightly) nasty explicit form

Explit form:

$$\log \rho_\theta(Y_k | Y_0^{l-1}) = \log \int_{\mathcal{X}} \gamma_\theta(x_k, Y_k) Q_\theta(x_{k-1}, dx_k) P_\theta(X_{k-1} \in dx_{k-1} | Y_0^{k-1})$$

γ_θ transition density, i.e.

$$\int_{\mathcal{X}} \int_{\mathcal{Y}} \mathbb{1}_C(x', y') \gamma_\theta(x', y') \phi(dy') Q_\theta(x, dx') = \int_{\mathcal{X}} \int_{\mathcal{Y}} \mathbb{1}_C(x', y') G_\theta(x', dy') Q_\theta(x, dx')$$

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- Say $\gamma_\theta(x, y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-y)^2}{2\sigma^2})$ and $\mathcal{X}$ is a finite state space $\mathcal{X} = \{\mu_1, \dots, \mu_d\}$.
- ... or say $\mathcal{X}$ is a compact manifold of dimension k, Q has diffusion type transition density $q^{(t_0)}(x, x') \sim (4\pi t_0)^{-k/2} e^{-d(x,x')^2/(4t_0)}$
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$\rightsquigarrow$ **We don't do this!**

What one usually does in the Hidden Markov case

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- Remember: $n^{-1} \sum_{k=0}^{n-1} g(\tau^k) \to \mathrm{E}[g]$ almost surely, where τ is the shift operator of a measure-preserving dynamical system.
- Example: $\tau^k \rightsquigarrow (\dots, Y_{k-1}, Y_k)$
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But if homogeneity fails?

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- The above introduced *finite memory approximation* is in the spirit of local stationarity.
- *infinite memory approximation* is clearly off the table.

Treating with the likelihood function

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \log \rho_\theta(Y_{k,N} | Y_{t_N^-,N}^{k-1,N})$$

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$(bN)^{-1}\sum_{k=t_N^-}^{t_N^+}\log\rho_\theta(Y_{k,N}|Y_{t_N^-,N}^{k-1,N})$ **finite memory approximation**

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Treating with the likelihood function

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$$\rightsquigarrow (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N})$$

$$= (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) - \mathrm{E}\left[\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] \right)$$

$$+ (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\mathrm{E}\left[\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] - \mathrm{E}\left[\log \rho_\theta(Y_k(t) | Y_{k-l}^{k-1}(t)) \right] \right)$$

$$+ (bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \mathrm{E}\left[\log \rho_\theta(Y_k(t) | Y_{k-l}^{k-1}(t)) \right].$$

A LLN for triangular arrays

$$(bN)^{-1}\sum_{k=t_N^-}^{t_N^+}\left(\log \rho_\theta(Y_{k,N}|Y_{k-l,N}^{k-1,N}) - \mathrm{E}\left[\log \rho_\theta(Y_{k,N}|Y_{k-l,N}^{k-1,N})\right]\right)$$

Proposition

Let $\{\{Z_{k,N}\}_{0\leq k\leq N}\}_{N\geq 0}$ be a triangular array of random variables which is adapted to the family of filtrations $\{\mathbb{F}_N = (\mathcal{F}_{k,N})_{0\leq k\leq N}\}_{N\geq 0}$. Assume there is a C and a $1>\rho>0$ such that

$$|\,\mathrm{E}\,[Z_{k,N}|\mathcal{F}_{m,N}]\,| \leq C\cdot\rho^{k-m},$$

then for $bN\to\infty$

$$(bN)^{-1}\sum_{k=t_N^-}^{t_N^+} Z_{k,N} \xrightarrow{\mathrm{P}} 0.$$

How can we adapt a consistency proof?

$$(bN)^{-1} \sum_{k=t_N^-}^{t_N^+} \left(\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) - \mathrm{E}\left[\log \rho_\theta(Y_{k,N} | Y_{k-l,N}^{k-1,N}) \right] \right)$$

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Definition (Doeblin condition)

Let K be a Markov kernel from $(\mathcal{Z}_1, \mathcal{A}_{\mathcal{Z}_1})$ to $(\mathcal{Z}_2, \mathcal{A}_{\mathcal{Z}_2})$. If there is a probability measure μ on $(\mathcal{Z}_2, \mathcal{A}_{\mathcal{Z}_2})$ and an ε such that

$$K(x, A) \geq \varepsilon \mu(A) \text{ for all } A \in \mathcal{A}_{\mathcal{Z}_2},$$

we say that Q satisfies a Doeblin condition.

The quite powerful Doeblin condition

$$K(x, A) \geq \varepsilon\mu(A) \text{ for all } A \in \mathcal{A}_{\mathcal{Z}_2},$$
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Proposition (Doeblin condition leads to a contraction)

If $Q_1, \ldots, Q_n$ are Markov kernels on $(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ and satisfy the Doeblin condition with common ε, we get for every $\nu \in \mathcal{M}_0(\mathcal{X}, \mathcal{A}_{\mathcal{X}})$ the bound

$$\|Q_n \cdots Q_1\nu\|_{\mathrm{TV}} \leq (1-\varepsilon)^n \|\nu\|_{\mathrm{TV}}.$$

- Think of $\nu = \delta_x - \delta_y$ for $x, y \in \mathcal{X}$.
- Doeblin $\rightsquigarrow$ exponential forgetting of initial condition $\rightsquigarrow$ convergence against expected value

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The (Doeblin-powered) final appearance of local stationarity

$$(bN)^{-1}\sum_{k=t_N^-}^{t_N^+}\left(\mathrm{E}\left[\log\rho_\theta(Y_{k,N}|Y_{k-l,N}^{k-1,N})\right]-\mathrm{E}\left[\log\rho_\theta(Y_k(t)|Y_{k-l}^{k-1}(t))\right]\right) \qquad \left|\int f d\nu\right|\leq\|f\|_\infty\|\nu\|_{\mathrm{TV}}$$

Theorem ($\|\cdot\|_{\mathrm{TV}}$-local stationarity of Markov chains, [Tru19])

Let $\{K_u\}_{u\in[0,1]}$ be a family of Markov kernels on $(\mathcal{Z},\mathcal{B}(\mathcal{Z}))$ and let $\{\{Z_{k,N}\}_{0\leq k\leq N}\}_{N\geq 0}$ be a triangular array of Markov chains with transition kernels $\mathrm{P}(Z_{k,N}\in dx|Z_{k-1,N})=K_{\frac{k}{N}}(Z_{k-1,N},dx)$ that satisfies the following two conditions:

(A1) All transition kernels K_u satisfy a Doeblin condition with common constant $\varepsilon>0$.

(A2) $\sup_{z\in\mathcal{Z}}\|K_u(z,\cdot)-K_v(z,\cdot)\|_{\mathrm{TV}}\leq L|u-v|$

Then, for every l we have

$$\|\pi_{k,N}^{k+l,N}-\pi_{l+1}(t)\|_{TV}\leq C_{l+1}\left(\left|t-\frac{k}{N}\right|+\frac{1}{N}\right).$$

On possible assumptions

- We use $|\log \rho_\theta(Y_{k,N}|Y_0^k)| \leq C$ type bounds for total variation $\rightsquigarrow \kappa \leq \gamma_\theta \leq \kappa^{-1}$ for $\kappa \in (0,1)$.

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Theorem

Assume the transition kernels $T_{k,N}$ of the array of Hidden Markov models are given by $T_{\theta^\infty(k/N)}$ for $\theta^\infty : [0,1] \to \Theta$ Lipschitz. Assume further that the above assumptions hold. Then, the maximum likelihood estimator is (weakly) consistent.

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Identifiability is important – but it is a property of the stationary distributions.

The Doeblin condition and possible examples

- $\mathcal{X} = \{x_1, \ldots, x_n\}, \mathcal{Y} = \{y_1, \ldots, y_m\}$, $\inf_\theta \inf_{i,j\in\mathcal{X}} q_\theta(i,j) > 0$, $\kappa \leq \gamma_\theta \leq \kappa^{-1}$.
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- But even in simple finite state space models, Doeblin minorization is often not fulfilled: $\mathcal{X} = \{1, 2, 3\}$,

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- On the level of random variables,

$$f(x, X_{k-1}) = \begin{cases} V_\nu & \text{if } U \leq \varepsilon \\ g_K(X_{k-1}, W) & \text{if } U \geq \varepsilon \end{cases}$$

for $U, W \sim \mathcal{U}([0, 1])$ and $V_\nu \sim \nu$ independent and g_K suitable.

But what about the AR(1) case?

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  is also an interesting (locally stationary) Markov model. **Kernels do not satisfy a Doeblin condition.**
- The seminal paper [Dou+11] covered such AR(1) models (and many other models from classical TSA).
- Quite intrinsic usage of ergodic theorem, seems to intentionally circumvent invoking (stronger) forgetting properties.

Time series models and the notion of L^p-local stationarity

All is not lost!

Definition (L^p-local stationarity in the spirit of [Dah12])

A triangular array of Hidden Markov chains $\{\{(X_{k,N}, Y_{k,N})\}_{0\leq k\leq N}\}_{N\geq 0}$ is called L^p-locally stationary, if for every $u \in [0,1]$ there is a stationary Hidden Markov chain $\{(\widetilde{X}_k(u), \widetilde{Y}_k(u))\}_{k\in\mathbb{Z}}$ such that for every sequence k such that $\frac{k}{N} \to u$ we have $\|\varepsilon_{k,N}(u) := Y_{k,N} - Y_k(u)\|_p \to 0$.

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Bound the second term by an interpolation technique for each component.

$$|d| \leq |\varepsilon_{k,N}(t)| \int_0^1 |\partial_{y_k} \log \rho_\alpha^\nu(\widetilde{Y_k}(t) + r\varepsilon_{k,N}(t), \widetilde{Y_{k+1}}(t))|dr + |\varepsilon_{k+1,N}(t)| \int_0^1 |\partial_{y_{k+1}} \log \rho_\alpha^\nu(Y_{k,N}, \widetilde{Y_{k+1}}(t) + r\varepsilon_{k+1,N}(t))|dr.$$

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is a nice bound. Vanishes in L^1 by Hoelder and Fubini (in case of right integrability conditons), **if we can bound the derivative**.

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$$\begin{aligned}
&|\partial_{y_1} \log \rho_\alpha^\nu(y_1, y_2)| \\
=&\frac{\partial_{y_1} \int_{\mathcal{X}^2} \gamma(|y_2 - x_2|) \exp(-(x_2 - \alpha x_1)^2/2)\gamma(|y_1 - x_1|)\lambda(dx_2) q_\nu(x_1)\lambda(dx_1)}{\int_{\mathcal{X}^2} \gamma(|y_2 - x_2|) \exp(-(x_2 - \alpha x_1)^2/2)\gamma(|y_1 - x_1|)\lambda(dx_2) q_\nu(x_1)\lambda(dx_1)} \\
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Thank you for your attention!

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